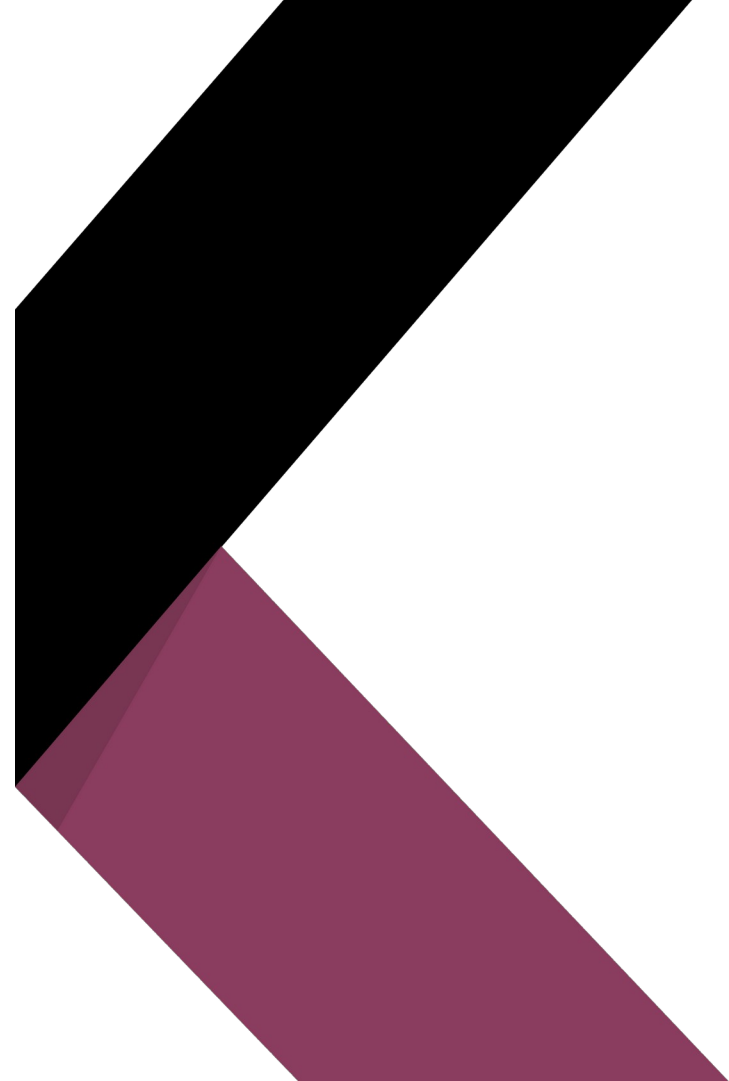


# Academy

# Ethernet

September 2022

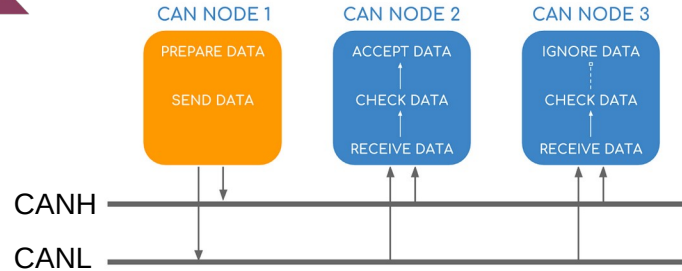




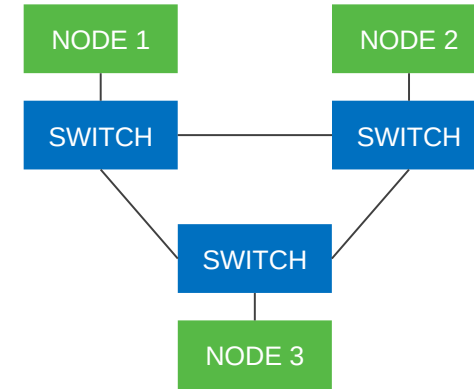
# Why Ethernet in automotive?



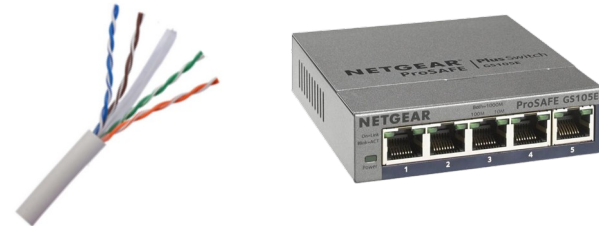
## CAN vs ETH networks



- ~8Mb/s
- **Single** of twisted pair copper wire
- All messages are **broadcast** messages
- **Signal** oriented (the i-th bit has a specific meaning)



- ~100Gb/s
- **Pairs** of twisted pair copper wire
- All messages have **source** and **destination** (they can be broadcast)
- **Service** oriented (send/require a msg)





# OSI Layers



# OSI Layers

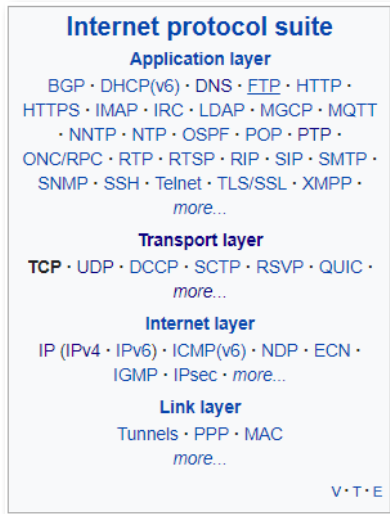
The **Open Systems Interconnection model (OSI model)** is a conceptual model that partitions the flow of data in a communication system into seven abstraction layers:

| Layer        |   |              | Protocol data unit (PDU) | Function                                                                                                                                      |
|--------------|---|--------------|--------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| Host layers  | 7 | Application  | Data                     | High-level APIs, including resource sharing, remote file access                                                                               |
|              | 6 | Presentation |                          | Translation of data between a networking service and an application; including character encoding, data compression and encryption/decryption |
|              | 5 | Session      |                          | Managing communication sessions, including the exchange of information in the form of multiple back-and-forth transmissions between two nodes |
|              | 4 | Transport    | Segment, Datagram        | Reliable transmission of data segments between points on a network, including segmentation, acknowledgement and multiplexing                  |
| Media layers | 3 | Network      | Packet                   | Structuring and managing a multi-node network, including addressing, routing and traffic control                                              |
|              | 2 | Data link    | Frame                    | Reliable transmission of data frames between two nodes connected by a physical layer                                                          |
|              | 1 | Physical     | Bit, Symbol              | Transmission and reception of raw bit streams over a physical medium                                                                          |

**Not used by Ethernet/IP**

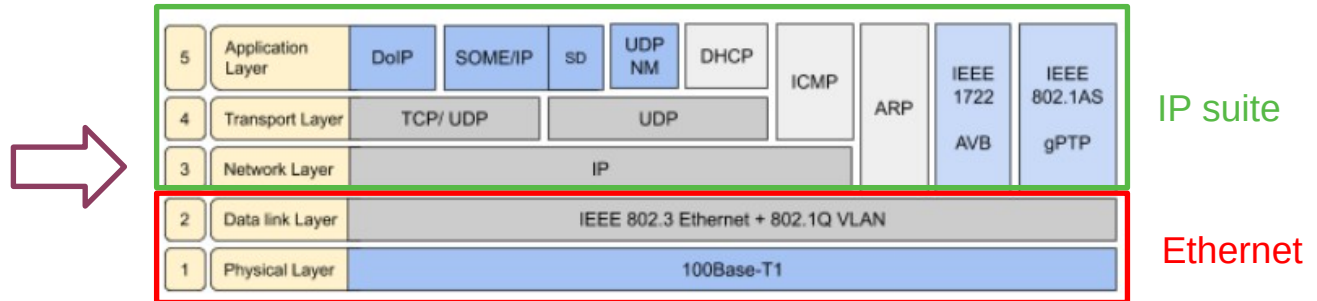


# Ethernet



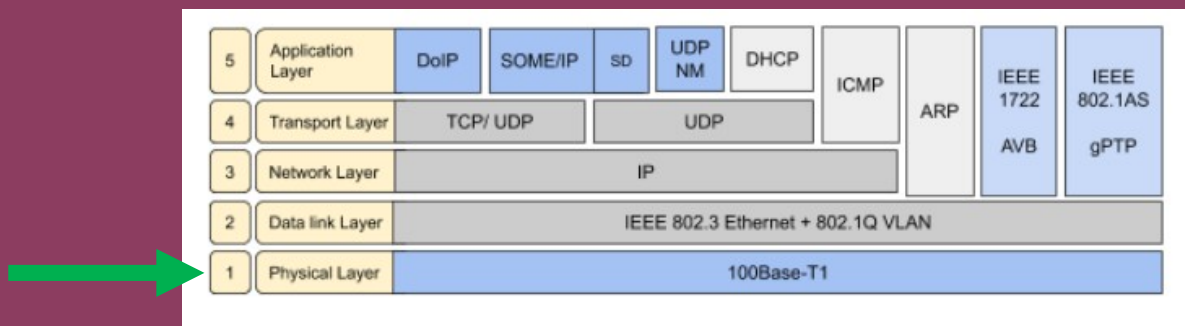
There are many protocols developed at different layers of the OSI model.

Here we will look at only the ones used in automotive:



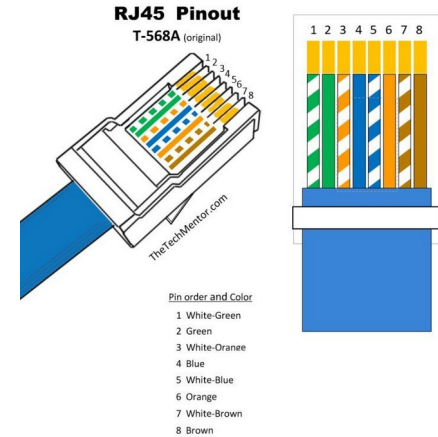


# Physical Layer





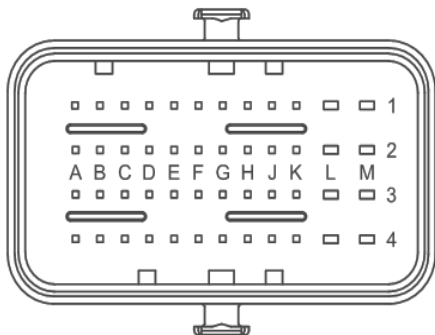
## Physical Layer



Many speeds from 1 Mbit/s to 400 Gbit/s  
Can support multiple speeds by using **autonegotiation**



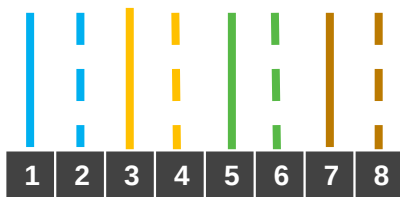
## Physical Layer - Example



| Cavity Number | Signal Name |
|---------------|-------------|
| A1            | ETH-RXP     |
| A2            | ETH-TXP     |
| B1            | ETH-RXM     |
| B2            | ETH-TXM     |

This configuration: 100Base-T

Let's play! Which wires connect to what?



| Pin | Name   | Description          | Cable Color  |
|-----|--------|----------------------|--------------|
| 1   | TX_D1+ | Tranceive Data+      | white/green  |
| 2   | TX_D1- | Tranceive Data-      | green        |
| 3   | RX_D2+ | Receive Data+        | white/orange |
| 4   | BI_D3+ | Bi-directional Data+ | blue         |
| 5   | BI_D3- | Bi-directional Data- | white/blue   |
| 6   | RX_D2- | Receive Data-        | orange       |
| 7   | BI_D4+ | Bi-directional Data+ | white/brown  |
| 8   | BI_D4- | Bi-directional Data- | brown        |



## What does it mean?

|    | $2^7$ | $2^6$ | $2^5$ | $2^4$ | $2^3$ | $2^2$ | $2^1$ | $2^0$ |
|----|-------|-------|-------|-------|-------|-------|-------|-------|
| 19 | 0     | 0     | 0     | 1     | 0     | 0     | 1     | 1     |

LSB

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## Physical Layer – Network topologies

1 Point to point



2 Bus



3 Ring



4 Star



5 Tree



6 Mesh

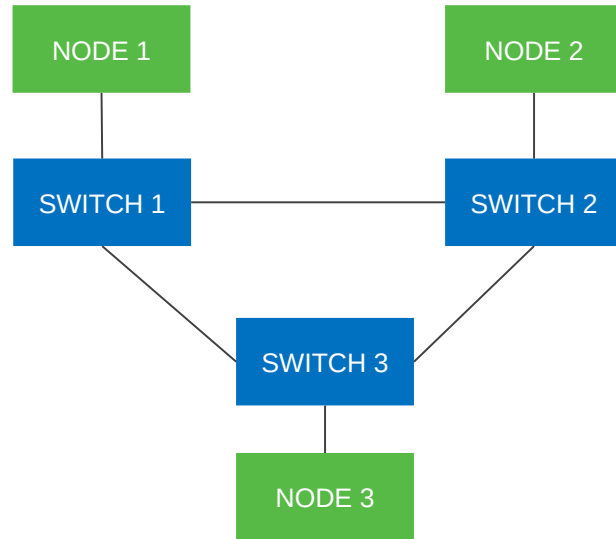


7 Hybrid

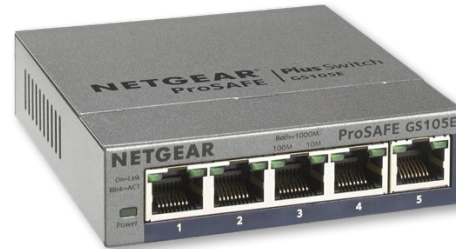




## Physical Layer – Switched Network



General Purpose



Automotive  
directly on pcb



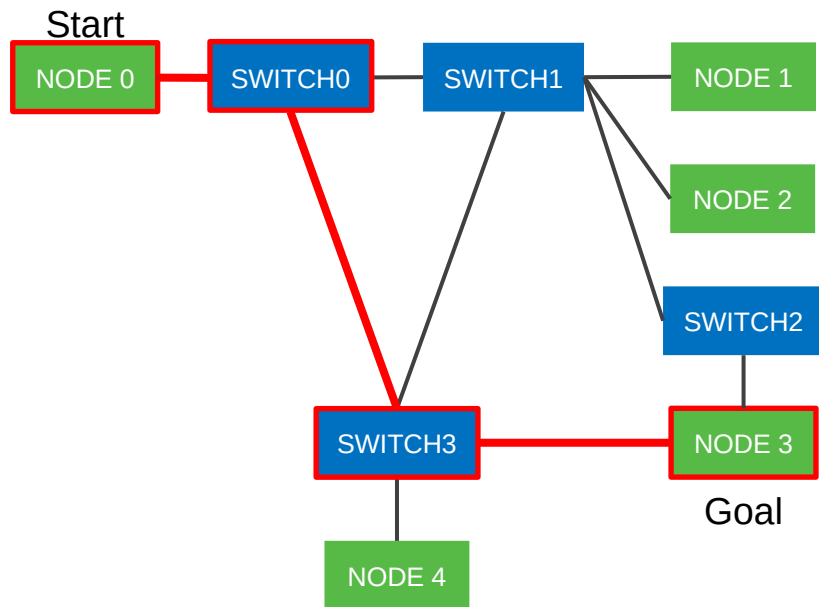
Switch ports can be:

- **Access** port (untagged port)
- **Trunking** Port (tagged port)
- **Allowing** port

Aside from the SGW, generally **all** switch ports in automotive are trunking ones.



## Physical Layer – Switched Network Example



Node 0 wants to send a message to Node 3

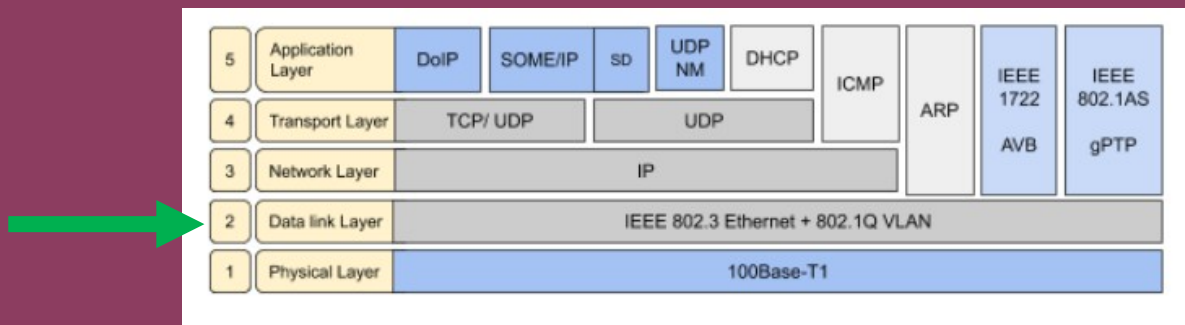
What path does it takes?

Keep in mind:

- Every switch knows what nodes are connect to it
- At network setup/startup, all switches construct their *Shortest Spanning Tree*
- Node0 uses the Node3 Mac address as destination



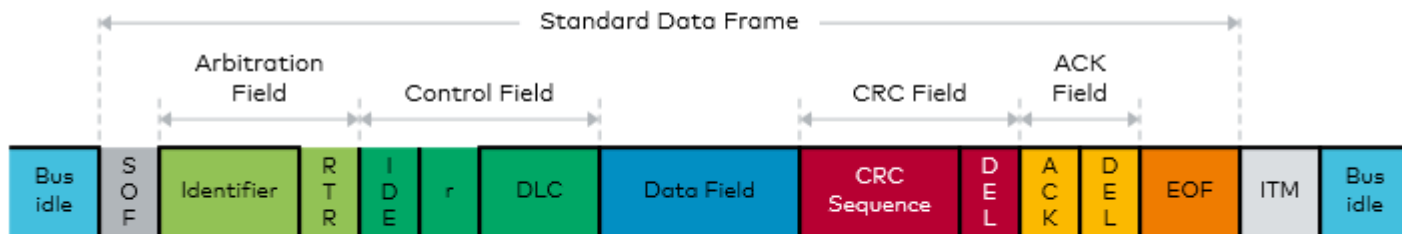
# Data Link Layer - Frame



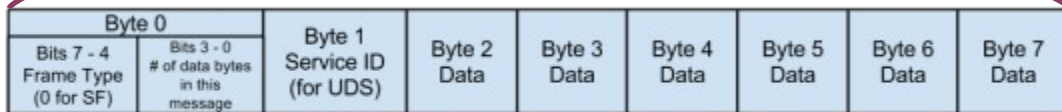


## Example of "Encapsulation" on CAN

CAN  
Frame

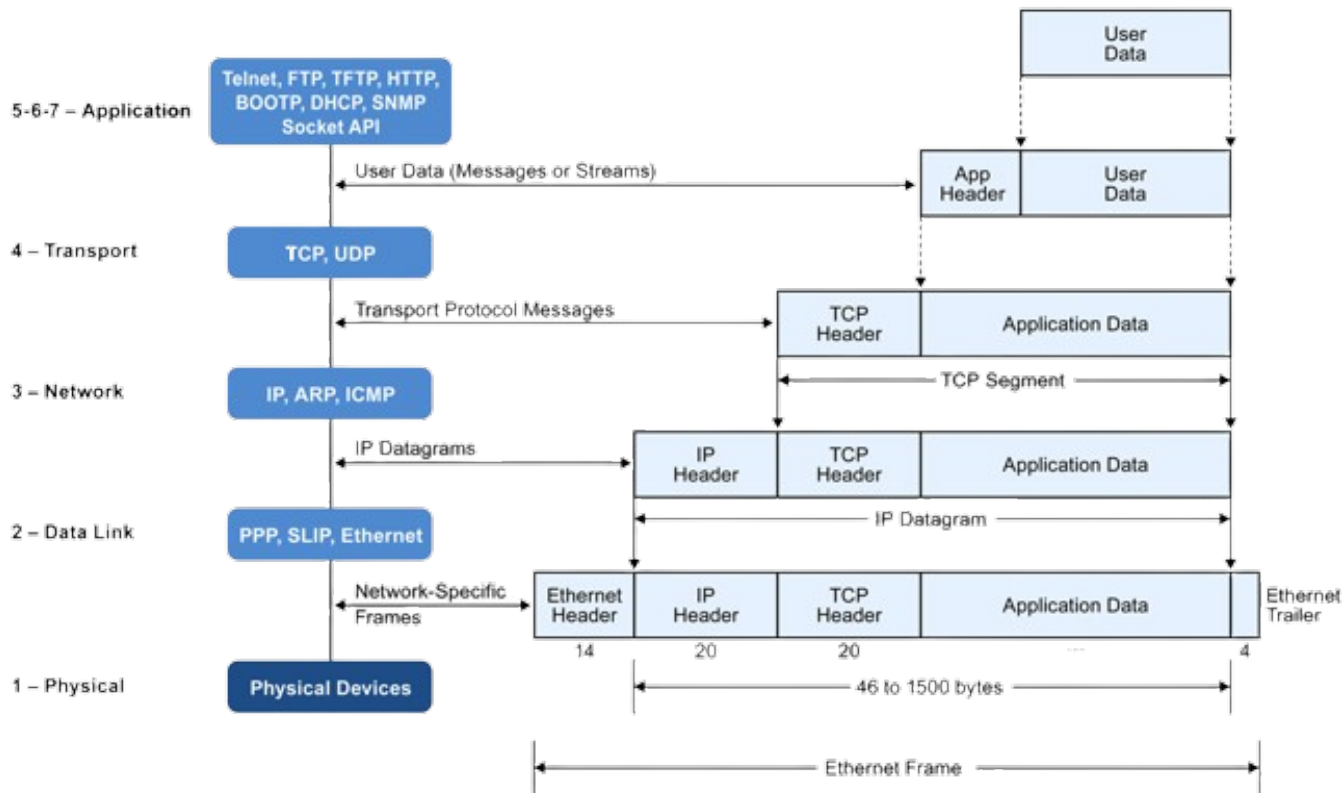


UDS SF





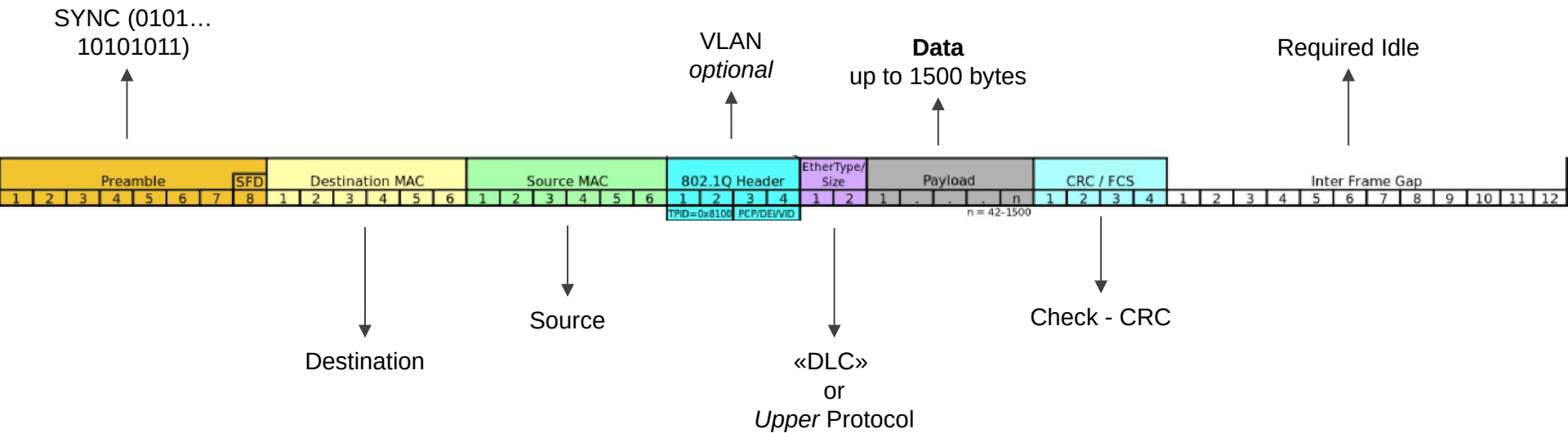
# Encapsulation on ethernet





# Ethernet Frame

|          |     |                         |                    |           |         |     |
|----------|-----|-------------------------|--------------------|-----------|---------|-----|
| Preamble | SFD | Destination MAC Address | Source MAC Address | EtherType | Payload | FCS |
|----------|-----|-------------------------|--------------------|-----------|---------|-----|



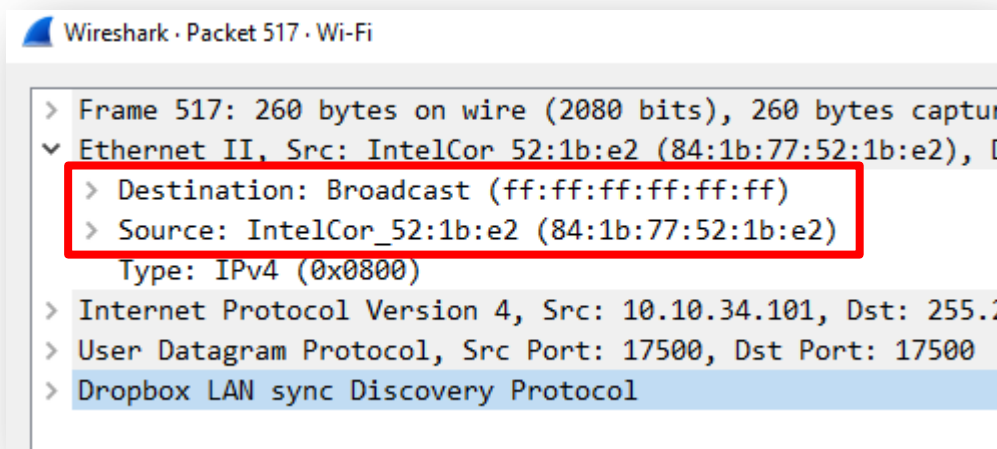


## Media Access Control (MAC) address

Media Access Control address (MAC) is a **unique** and **mandatory** identifier assigned to a Network Interface Controller (NIC).

6 bytes:

- 3 bytes company who produced the NIC
- 3 bytes the specific ID



Other companies [link](#):

70 8B CD - Asus

EC FA 03 - FCA

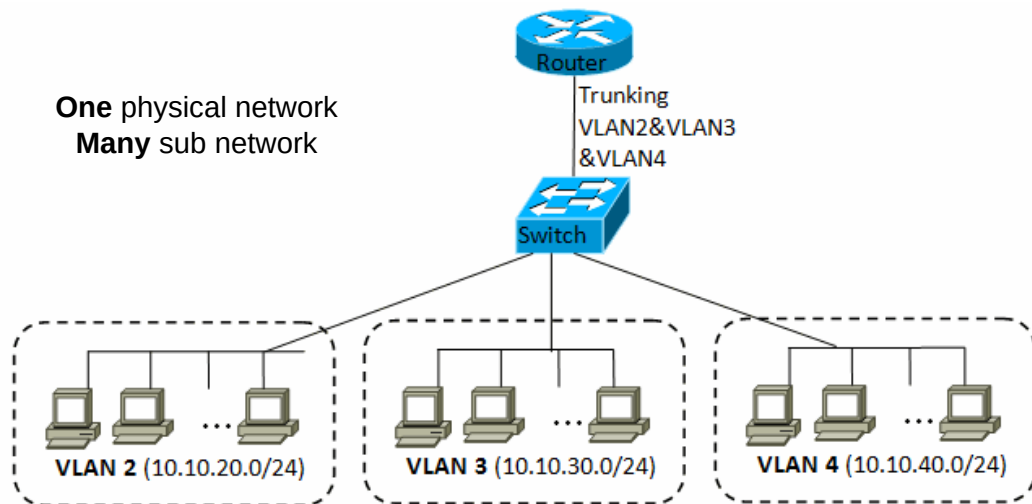
E8 FC AF - NetGear

|          |     |                               |                          |           |         |     |
|----------|-----|-------------------------------|--------------------------|-----------|---------|-----|
| Preamble | SFD | Destination<br>MAC<br>Address | Source<br>MAC<br>Address | EtherType | Payload | FCS |
|----------|-----|-------------------------------|--------------------------|-----------|---------|-----|



## Virtual Local Access Network (VLAN) – Optional

|          |     |                               |                          |           |         |     |
|----------|-----|-------------------------------|--------------------------|-----------|---------|-----|
| Preamble | SFD | Destination<br>MAC<br>Address | Source<br>MAC<br>Address | EtherType | Payload | FCS |
|----------|-----|-------------------------------|--------------------------|-----------|---------|-----|



| VID | Domain                |
|-----|-----------------------|
| 6   | ADAS Integration      |
| 102 | ADAS Data Fusion      |
| 103 | Positioning Data      |
| 104 | Object Detection      |
| 105 | Trajectory            |
| 107 | Ultrasonic Near Range |
| 106 | HMI Application       |



# EtherType

|          |     |                               |                          |           |         |     |
|----------|-----|-------------------------------|--------------------------|-----------|---------|-----|
| Preamble | SFD | Destination<br>MAC<br>Address | Source<br>MAC<br>Address | EtherType | Payload | FCS |
|----------|-----|-------------------------------|--------------------------|-----------|---------|-----|

## Two Meanings

< 0x05DC (1500 dec)

> 0x05DC (1500 dec)

### Length Interpretation

Payload length, like DLC in CAN

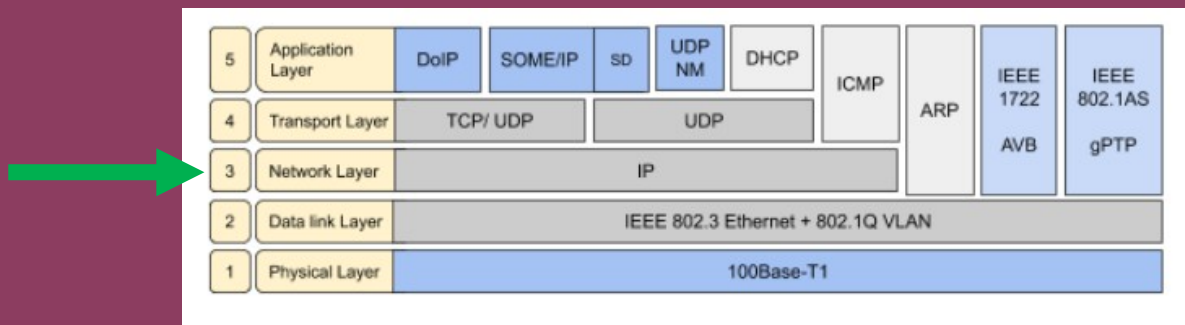
### Type Interpretation

Payload "Type"/Upper Protocol

- 0x0800, indicate a datagram IPv4
- 0x0806, indicate a frame ARP
- 0x22EA, Stream Reservation Protocol (SRP)
- 0x86DD, indicate a datagram IPv6
- 0x8808, Ethernet flow control
- 0x88F7, Precision Time Protocol

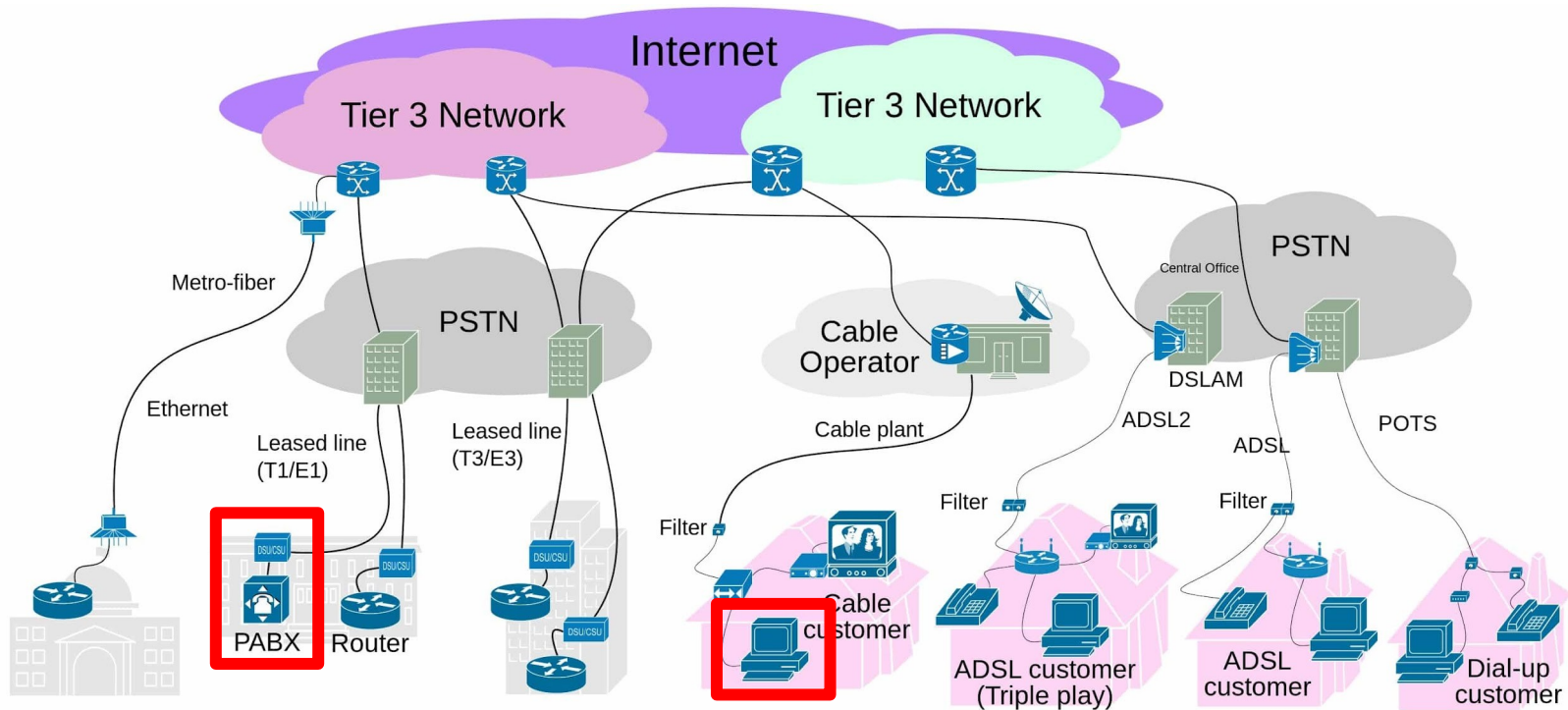


# Network Layer - Packet





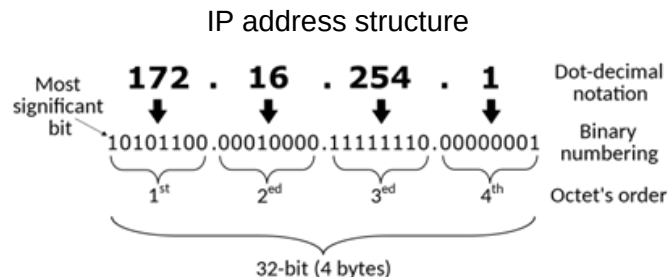
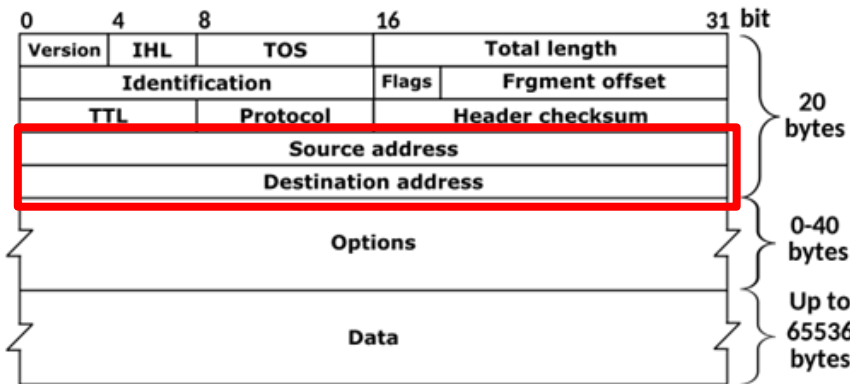
# Internet Protocol suite





# Internet Protocol suite

IP has the task of delivering packets from the source host to the destination host **solely** based on the IP addresses even **across** network boundaries



Usually expressed as quad-dotted IP address

192.0.2.235

1100 0000 . 0000 0000 . 0000 0010 . 1110 1011

3221226219 decimal



## Example1: IP 1/2

Enter a domain (e.g. google.com) in <https://who.is/>  
Go to DNS section, you will find the IP address of that domain

For example I searched google's IP, and I got:

172.253.63.139

In Wireshark set a filter (the green bar) in order to only see the traffic to and from that IP address:  
`ip.addr == 172.253.63.139`

In Command Prompt and type:

`ping 172.253.63.139`

```
C:\Users\Kineton>ping 172.253.63.139

Esecuzione di Ping 172.253.63.139 con 32 byte di dati:
Risposta da 172.253.63.139: byte=32 durata=136ms TTL=105
Risposta da 172.253.63.139: byte=32 durata=137ms TTL=105
Risposta da 172.253.63.139: byte=32 durata=140ms TTL=105
Risposta da 172.253.63.139: byte=32 durata=141ms TTL=105
```

| No.  | Time      | Source         | Destination    | Protocol | Length | Info                |
|------|-----------|----------------|----------------|----------|--------|---------------------|
| 1808 | 64.932459 | 10.10.34.15    | 172.253.63.139 | ICMP     | 74     | Echo (ping) request |
| 1809 | 65.070399 | 172.253.63.139 | 10.10.34.15    | ICMP     | 74     | Echo (ping) reply   |
| 1813 | 65.955977 | 10.10.34.15    | 172.253.63.139 | ICMP     | 74     | Echo (ping) request |
| 1814 | 66.092539 | 172.253.63.139 | 10.10.34.15    | ICMP     | 74     | Echo (ping) reply   |
| 1846 | 66.968892 | 10.10.34.15    | 172.253.63.139 | ICMP     | 74     | Echo (ping) request |
| 1850 | 67.106724 | 172.253.63.139 | 10.10.34.15    | ICMP     | 74     | Echo (ping) reply   |
| 1879 | 67.990780 | 10.10.34.15    | 172.253.63.139 | ICMP     | 74     | Echo (ping) request |
| 1880 | 68.127596 | 172.253.63.139 | 10.10.34.15    | ICMP     | 74     | Echo (ping) reply   |

Who is 10.10.34.15?

```
C:\Users\Kineton>ipconfig /all
```

#### Configurazione IP di Windows

```
Nome host . . . . . : L-FranPetracci
Suffisso DNS primario . . . . . :
Tipo nodo . . . . . : Ibrido
Routing IP abilitato. . . . . : No
Proxy WINS abilitato . . . . . : No
```

#### Scheda Ethernet Ethernet:

```
Stato supporto. . . . . : Supporto disconnesso
Suffisso DNS specifico per connessione: homenet.telecomitalia.it
Descrizione . . . . . : Realtek PCIe GbE Family Controller
Indirizzo fisico. . . . . : 70-8B-CD-23-88-39
DHCP abilitato. . . . . : Si
Configurazione automatica abilitata : Si
```

#### Scheda LAN wireless Connessione alla rete locale (LAN)\* 1:

```
Stato supporto. . . . . : Supporto disconnesso
Suffisso DNS specifico per connessione:
Descrizione . . . . . : Microsoft Wi-Fi Direct Virtual Adapter
Indirizzo fisico. . . . . : 76-C6-3B-A9-DD-39
DHCP abilitato. . . . . : Si
Configurazione automatica abilitata : Si
```

#### Scheda LAN wireless Connessione alla rete locale (LAN)\* 2:

```
Stato supporto. . . . . : Supporto disconnesso
Suffisso DNS specifico per connessione:
Descrizione . . . . . : Microsoft Wi-Fi Direct Virtual Adapter #2
Indirizzo fisico. . . . . : 86-C6-3B-A9-DD-39
DHCP abilitato. . . . . : Si
Configurazione automatica abilitata : Si
```

#### Scheda LAN wireless Wi-Fi:

```
Suffisso DNS specifico per connessione:
Descrizione . . . . . : Qualcomm Atheros QCA9377 Wireless Network Adapter
Indirizzo fisico. . . . . : 74-C6-3B-A9-DD-39
DHCP abilitato. . . . . : Si
Configurazione automatica abilitata : Si
Indirizzo IPv4 locale assegnato al collegamento : fe80::c610:c736:fe7:b8ee%21(Preferenziale)
Indirizzo IPv4 . . . . . : 10.10.34.15(Preferenziale)
Subnet mask . . . . . : 255.255.255.0
Lease ottenuto. . . . . : venerdì 23 settembre 2022 08:33:50
Scadenza lease . . . . . : sabato 24 settembre 2022 08:33:50
Gateway predefinito . . . . . : 10.10.34.1
Server DHCP . . . . . : 10.10.34.1
IAID DHCPv6 . . . . . : 292865595
DUID Client DHCPv6 . . . . . : 00-01-00-01-29-6F-71-8E-70-8B-CD-23-88-39
Server DNS . . . . . : 185.40.212.230
                  : 185.40.213.231
                  : 8.8.8.8
NetBIOS su TCP/IP . . . . . : Attivato
```

## Example1: IP 2/2

Who is 10.10.34.15? Is that us?

In cmd prompt type:

ipconfig /all

We find also the MAC address of my PC!

74-C6-3B-A9-DD-39

Let's open a request message (PC -> google)

Wireshark · Packet 1813 · Wi-Fi

```
> Frame 1813: 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on 0
  Ethernet II, Src: AzureWav_a9:dd:39 (74:c6:3b:a9:dd:39), Dst: Routerbo
    > Destination: Routerbo 93:5d:45 (dc:2c:6e:93:5d:45)
      Source: AzureWav_a9:dd:39 (74:c6:3b:a9:dd:39)
      Type: IPv4 (0x0800)
    > Internet Protocol Version 4, Src: 10.10.34.15, Dst: 172.253.63.139
    > Internet Control Message Protocol
```

It's our router!

It is NOT google MAC address!



## Example2: IP basic protocols

ECU1 has a packet to send to ECU2

Through [DNS](#), it determines that ECU2 has the IP address 192.168.0.55.

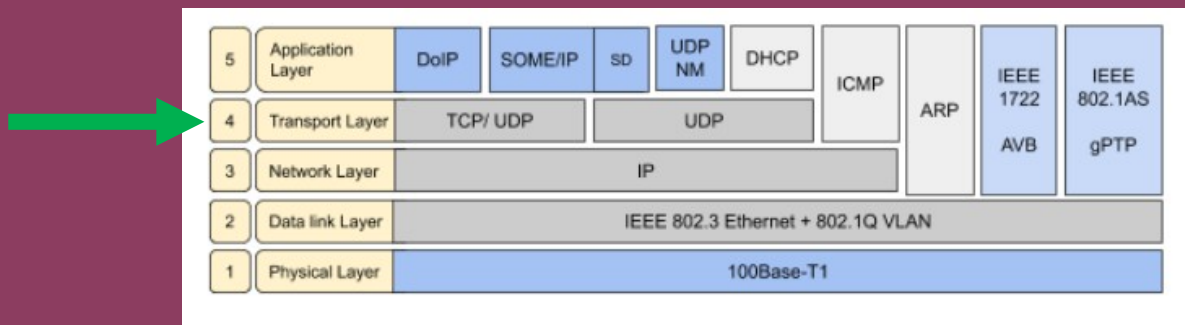
1. ECU1 uses a cached [ARP](#) table. Does not find MAC address associated to 192.168.0.55
2. ECU1 sends an ARP broadcast frame requesting an answer for 192.168.0.55
3. ECU2 responds with an ARP message
4. ECU1 receives and caches the response information in its ARP table
5. ECU1 can now send the message

Mac Addresses

| Source            | Destination       | Protocol |
|-------------------|-------------------|----------|
| 80:61:5f:0c:27:61 | ff:ff:ff:ff:ff:ff | ARP      |
| ec:fa:03:00:e8:11 | 80:61:5f:0c:27:61 | ARP      |



# Transport Layer



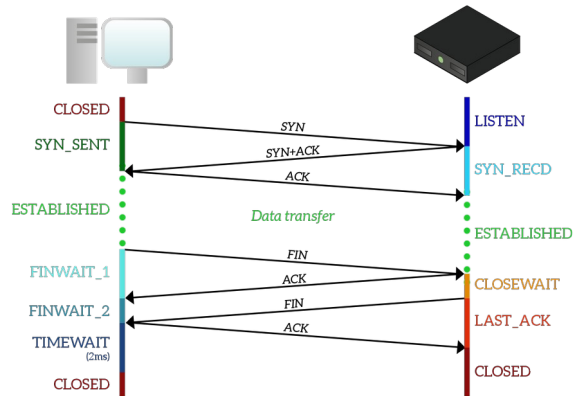


# Transport Protocols

## Transmission Control Protocol (TCP)

- **Robust**
- high latency (seconds)

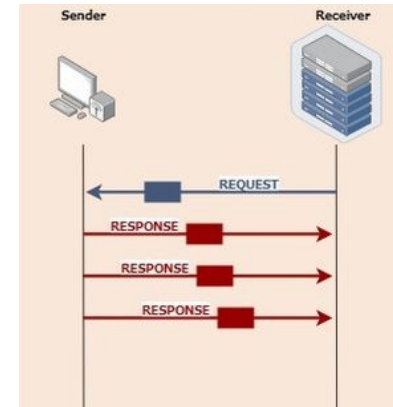
E.g. Diagnostics



## User Datagram Protocol (UDP)

- minimum protocol mechanisms
- **low** latency

E.g. Streaming audio/video





## Meaningful Joke!



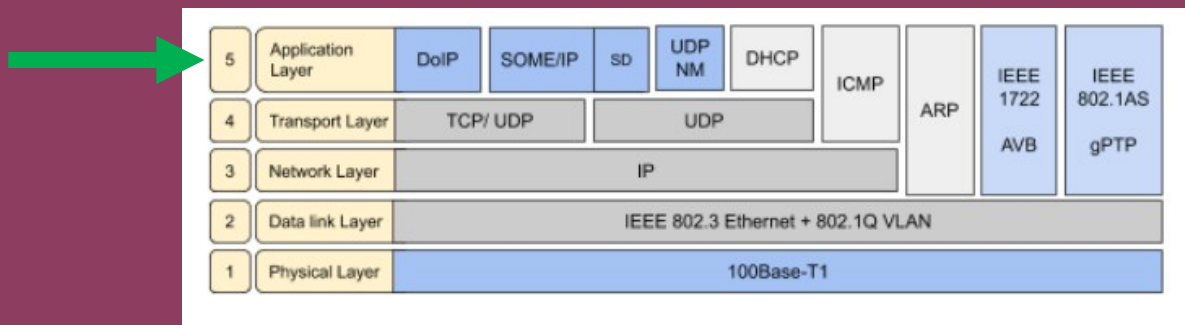


## TCP/UDP Guidelines

- Use TCP only if very large chunks of data need to be transported (> 1400 Bytes) and no hard latency requirements in the case of errors exists
- Use UDP if very hard latency requirements (<100ms) in case of errors is needed
- Use UDP together with SOME/IP-TP if very large chunks of data need to be transported (> 1400 Bytes) and hard latency requirements in the case of errors exists
- Try using external transport or transfer mechanisms (Network File System, APIX link, 1722, ...) when they are more suited for the use case. In this case SOME/IP can transport a file handle or a comparable identifier. This gives the designer additional freedom (e.g. in regard to caching).



# Application Layer

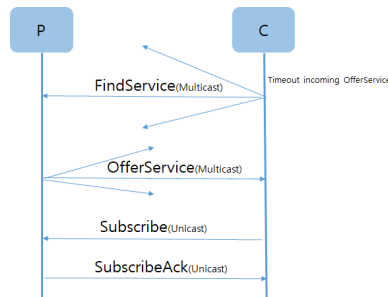
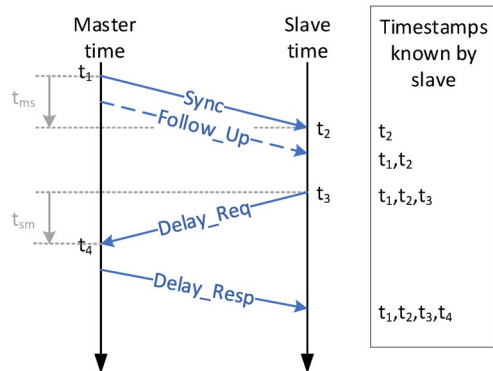




## Application Layer

These are topics for future lessons!

Some of protocols used in automotive:  
FOTA, AVB, gPTP, DOiP, SOME/IP, SOAD ...



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# Extras